Liquid Vector Spaces as a Target Category for Topological Quantum Field Theories

A Formally Verified Categorical Foundation

Research Proposal, March 2026

Abstract

We propose liquid vector spaces (Clausen–Scholze) as a replacement target category for topological quantum field theories in the infinite-dimensional regime. The standard target, topological vector spaces, fails to be abelian, which breaks the homological algebra required for TQFT gluing axioms. Condensed abelian groups, and their liquid subcategory, fix this completely: they form an abelian category with an exact completed tensor product.

We report on a formal verification effort using Lean 4 and Mathlib (v4.28.0), assisted by Harmonic's Aristotle system, establishing the categorical foundations for this program. The central result, that condensed abelian groups form a symmetric monoidal category, is proved *without any sorries or custom axioms*, assembling existing Mathlib infrastructure from sheaf theory, localization, enriched categories, and monoidal closed structure. We further prove, in full generality, that any TQFT valued in a braided monoidal subcategory transfers to the condensed setting along braided monoidal functors.

The remaining components (the monoidal embedding of Banach spaces into condensed abelian groups, exactness of the liquid tensor product, and formalized cobordism categories) are identified as concrete next steps, with precise Mathlib API gaps documented.

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1. Introduction

1.1 The Problem

A topological quantum field theory, in Atiyah's formulation, is a symmetric monoidal functor Z : Cob(d+1) → C from a cobordism category to some symmetric monoidal target category C. When C is the category of finite-dimensional vector spaces, the theory is clean and well-understood. Dijkgraaf–Witten, Reshetikhin–Turaev, and Turaev–Viro all live here comfortably.

But the physically interesting theories refuse to stay finite-dimensional. Chern–Simons theory with non-compact gauge group $SL(2,\mathbb{C})$, quantum gravity partition functions, anything involving honest path integrals over infinite-dimensional moduli. These force the state spaces $Z(\Sigma)$ assigned to boundary manifolds to be infinite-dimensional. The natural target becomes topological vector spaces. And here the trouble starts.

The category TopAb of topological abelian groups is **not abelian**. A continuous bijective homomorphism need not have a continuous inverse. This means a morphism can be simultaneously monic and epic without being an isomorphism. This is a fundamental failure that propagates through the entire homological algebra machinery. Exact sequences break. Derived functors misbehave. And crucially for TQFT: the gluing axiom, which says that cutting a manifold along a codimension-1 surface decomposes the state space via an exact sequence, becomes ill-defined.

The completed tensor product in TopVect compounds the problem. It is not exact: tensoring a short exact sequence with a topological vector space can destroy exactness. Since the TQFT tensor product (disjoint union of boundaries ↔ tensor product of state spaces) requires this operation, infinite-dimensional TQFTs built on TopVect are categorically deficient in a precise, formal sense.

1.2 The Proposed Solution

Clausen and Scholze's condensed mathematics provides what appears to be the right fix. The category of condensed abelian groups (sheaves of abelian groups on the pro-étale site of a point, equivalently on the coherent site of compact Hausdorff spaces) is an abelian category. Every morphism has a kernel and cokernel. Every monic epic is an isomorphism. The category has all limits and colimits.

More importantly, liquid vector spaces (a refinement of condensed modules) admit an exact completed tensor product. This is the content of the Liquid Tensor Experiment, Lean-verified by Commelin and Scholze in 2022. The exactness of this tensor product is precisely the property that the classical completed tensor product on topological vector spaces lacks, and precisely the property that TQFT gluing requires.

This proposal investigates the formal viability of replacing TopVect with LiqVect (or more generally, Cond(Ab)) as the target category for infinite-dimensional TQFTs. We don't just argue the case mathematically; we verify it in Lean 4.

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2. Formal Verification Results

All results in this section are formally verified in Lean 4 against Mathlib v4.28.0, using Harmonic's Aristotle system for assisted theorem proving. The codebase consists of three files totaling approximately 800 lines of Lean, with the final construction being entirely sorry-free.

2.1 CondensedAb Is a Symmetric Monoidal Category

Theorem (formally verified). The category CondensedAb of condensed abelian groups admits a symmetric monoidal structure via the sheafified pointwise tensor product. The sheafification functor is monoidal and braided.

The construction proceeds through four layers of existing Mathlib infrastructure, assembled in MonoidalViaLocalization.lean:

1. **Presheaf monoidal structure.** The functor category $\mathrm{CompHaus}^{op} \to \mathrm{Ab}$ inherits a pointwise monoidal structure from the tensor product on $\mathbb{Z}$-modules (Mathlib.CategoryTheory.Monoidal.FunctorCategory). Verified by inferInstance.
2. **Sheafification as localization.** The functor presheafToSheaf is a localization functor for the class J.W of local isomorphisms (Mathlib.CategoryTheory.Sites.Localization). Verified by inferInstance.
3. **W is a monoidal morphism property.** The class J.W is closed under tensoring. This is the key mathematical fact, proved in Mathlib via the internal hom: if $g \in W$ and F is any presheaf, then for any sheaf H, the tensor–hom adjunction gives $\mathrm{Hom}(F \otimes G_2, H) \cong \mathrm{Hom}(G_2, [F, H])$, and [F, H] is a sheaf, so $g \in W$ gives the required bijection. Instance: GrothendieckTopology.W.monoidal.
4. **Localized monoidal structure.** The general localization machinery constructs a monoidal category on any localization of a monoidal category at a monoidal morphism property. Applying this yields MonoidalCategory CondensedAb, plus braided and symmetric refinements.

The entire proof is six lines of Lean:

```
instance condensedAb_W_isMonoidal :
    ((coherentTopology CompHaus).W
      (A := ModuleCat (ULift ℤ))).IsMonoidal :=
  GrothendieckTopology.W.monoidal

instance condensedAb_monoidalCategory :
    MonoidalCategory CondensedAb :=
  Sheaf.monoidalCategory _ _
```

2.2 TQFT Transfer Theorem

Theorem (formally verified). If Z : S → C is a TQFT (braided monoidal functor from a cobordism-like category) and F : C → D is a braided monoidal functor, then Z ⋙ F : S → D is a TQFT.

Note: The formalization works at the level of braided monoidal functors, which is strictly more general than the symmetric monoidal functors of Atiyah's original definition. Since every symmetric monoidal category is braided, the theorem applies a fortiori to Atiyah-style TQFTs. The broader setting also covers non-symmetric examples relevant to 3-dimensional TQFTs (e.g., Reshetikhin-Turaev invariants from modular tensor categories).

This is proved abstractly with no sorries, no custom axioms, purely from Mathlib's typeclass inference on monoidal and braided functor composition.

Conditional application. The transfer theorem implies that if a braided monoidal functor Ban → CondensedAb were established (sending the projective tensor product to the condensed tensor product), then any Banach TQFT would automatically yield a condensed TQFT. This embedding is not yet formalized; see Section 3.2.1.

2.3 The Abelianness Advantage

Theorem (formally verified). In any abelian category (hence in CondensedAb), a morphism that is both monic and epic is an isomorphism.

This one-liner from Mathlib (`isIso_of_mono_of_epi`) formalizes the exact obstruction that makes TopAb fail as a TQFT target. The fact that this property holds in CondensedAb and fails in TopAb is the core categorical argument for the proposed replacement.

The file also records the abstract categorical fact that short exact sequences in abelian categories provide the structure needed for TQFT gluing (decomposition into mono, epi, and exactness at the middle term). These are structural observations that take exactness as a hypothesis, not proofs that the condensed tensor product *is* exact. The substantive analytic claim, that the liquid tensor product preserves exact sequences, depends on the Liquid Tensor Experiment and is not verified in this codebase. See Section 3.2.2.

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3. The Liquid TQFT Research Program

3.1 What Is Established

Property	Status	Source

CondensedAb is abelian	✅ Verified	Mathlib inferInstance
Mono + epi = iso (fails in TopAb)	✅ Verified	isIso_of_mono_of_epi
Symmetric monoidal structure	✅ Verified	Sheaf.monoidalCategory
Sheafification is monoidal	✅ Verified	Sites.Monoidal
TQFT transfer along monoidal functors	✅ Verified	AbstractTQFT.transfer
Gluing structure from abelian exactness	✅ Structural	ShortComplex.ShortExact
SemiNormedGrp → CondensedAb functor	✅ Verified	BanachEmbedding.lean
Embedding is faithful	✅ Verified	Constant functions at PUnit
Embedding is full (SemiNormedGrp)	❌ Disproved	Counterexample: c_0(N,Z)
Embedding preserves finite limits	⚠ 2 sorry stubs	Strategy documented
CommFrobeniusAlgebra formalized	✅ Verified	Cob2.lean
2d cobordism category (Cob2)	✅ Verified	Quotient of free morphisms
Trivial Frobenius on unit object	✅ Verified	coherence tactic
Ban → CondensedAb monoidal	⚠ Blocked	Needs projective tensor product
Liquid tensor product exact	⚠ Not ported	LTE (archived)

3.2 What Remains

3.2.1 The Embedding SemiNormedGrp → CondensedAb. The functor sending a seminormed abelian group V to the condensed abelian group S ↦ C(S, V) is now fully constructed and sorry-free, including the sheaf condition. Faithfulness is proved. However, formal verification revealed that **fullness is mathematically false** for general seminormed groups: a counterexample (summation on eventually-zero integer sequences) shows that continuous additive maps between seminormed abelian groups need not be bounded. Fullness holds when restricted to normed vector spaces over a nontrivially normed field (R or C), where continuous linearity implies boundedness. The correct fully faithful embedding therefore requires defining `NormedSpaceCat R` and restricting accordingly. The monoidal property (sending projective tensor products to condensed tensor products) remains blocked by the absence of projective tensor products in Mathlib.

Mathlib status: The functor exists (BanachEmbedding.lean). Preservation of finite limits has 2 sorry stubs with documented proof strategies. No projective tensor product in Mathlib.

3.2.2 Exactness of the Liquid Tensor Product. The central analytic result from the Liquid Tensor Experiment. Exists in an archived Lean 4 repository, not upstreamed to Mathlib. Difficulty: Hard.

3.2.3 Cobordism Categories. A combinatorial model of the 2d cobordism category is now formalized (Cob2.lean), including the full commutative Frobenius algebra structure that classifies 2d TQFTs. The category is defined as a quotient of free morphisms by Frobenius relations. A trivial Frobenius algebra on the monoidal unit object is proved sorry-free. The functor from Frobenius data to Cob2 has one sorry (tedious case analysis). Smooth cobordism categories remain unformalized in any proof assistant.

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4. Why Liquid Vector Spaces

Chern–Simons with non-compact gauge group. SL(2,ℂ) Chern–Simons theory produces infinite-dimensional state spaces where gluing requires exact sequences of distribution spaces, exactly where TopVect fails and LiqVect succeeds.

Factorization algebras. Costello–Gwilliam's framework relies on nuclear Fréchet spaces and completed tensor products. Liquid vector spaces provide an exact tensor product in a well-behaved abelian category, simplifying factorization products and observables.

Perturbative invariants. Configuration space integrals in perturbative Chern–Simons produce distributional forms whose differentials involve products of distributions. The condensed framework offers cleaner multiplicative structure.

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5. Methodology and Tooling

The formal verification was conducted in six iterative rounds using Harmonic's Aristotle system against Mathlib v4.28.0.

Round 1 (LiquidTQFT.lean): Established the abstract TQFT framework, verified CondensedAb is abelian, proved TQFT transfer. Identified monoidal structure as key missing piece, declared as axiom.

Round 2 (CondensedMonoidal.lean): Attempted direct construction. Built complete MonoidalCategoryStruct, proved 5/10 axioms. Reduced remaining axioms to two sheafification comparison isomorphisms.

Round 3 (MonoidalViaLocalization.lean): Discovered Mathlib.CategoryTheory.Sites.Monoidal already contains everything. Six lines, zero sorries.

Round 4 (BanachEmbedding.lean): Audited Mathlib for connections between normed spaces and condensed math (finding: none existed). Constructed the embedding functor SemiNormedGrp -> CondensedAb sorry-free, including the sheaf condition. First bridge between these library components.

Round 5 (BanachEmbedding.lean, continued): Proved faithfulness. Discovered that fullness is *mathematically false* for seminormed groups via a concrete counterexample. Constructed finite products for SemiNormedGrp (not previously in Mathlib). Reduced finite limit preservation to two sorry stubs.

Round 6 (Cob2.lean): Formalized commutative Frobenius algebras using Mathlib's Mon_/Comon_ infrastructure. Built the combinatorial 2d cobordism category as a quotient of free morphisms. Proved the trivial Frobenius algebra on the monoidal unit (all 9 axioms via coherence). Defined TQFT2d mirroring the abstract framework.

The iterative process produced both expected results and genuine mathematical surprises. The Session 3 discovery (monoidal structure already in Mathlib) and the Session 5 counterexample (fullness fails) were both findings that reshaped the project's direction. The final codebase is approximately 1280 active lines of Lean with 3 sorries and 0 axioms.

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6. Conclusion

We have formally verified that condensed abelian groups form a symmetric monoidal abelian category suitable as a TQFT target, and proved that any TQFT valued in a braided monoidal subcategory transfers to this setting. Beyond the categorical foundation, we constructed the first embedding functor from seminormed groups into condensed abelian groups (bridging two previously disjoint Mathlib libraries), formalized the commutative Frobenius algebra structure that classifies 2d TQFTs, and built a combinatorial 2d cobordism category. The formal verification also produced a genuine mathematical finding: the embedding from seminormed groups is faithful but not full, with fullness requiring restriction to normed vector spaces over a valued field.

The broader vision is a "liquid TQFT" framework where infinite-dimensional state spaces live in LiqVect rather than TopVect, with exact tensor products replacing the categorically broken completed tensor product of classical functional analysis. The remaining barriers are increasingly specific: projective tensor products in Mathlib (for the monoidal embedding), porting the LTE exactness result (for full gluing), and smooth cobordism formalization (for geometric TQFTs). The most promising near-term mathematical application remains reformulating Costello-Gwilliam factorization algebras in the condensed setting.

Six rounds of AI-assisted formal verification, starting from a blank slate and ending with approximately 1280 lines of active Lean across 5 files, 3 sorries, and 0 axioms. The process produced both expected results and surprises. The formal verification reported here is evidence that the condensed TQFT direction is mathematically sound, precisely scoped, and ready for deeper development.